

OPENPHARMASTABILITY REPORT

Stability analysis

ICH Q1E inspired decision support for **assay** at **25C/60RH**. The Python engine owns the model, confidence bound, crossing estimate, reproducibility metadata, and artifact record.

SUPPORTED SHELF LIFE

17 months

Statistical crossing 18.0 months. Rounded down to a whole month.

ATTRIBUTE

assay

CONDITION

25C/60RH

DIRECTION

decreasing

TERM

shelf life for product

Decision Summary

The headline decision is the earliest supported period implied by the selected model and one sided confidence bound. Internal ids are retained in badges for audit traceability.

SUPPORTED SHELF LIFE

17 months

STATISTICAL CROSSING

18.0 months

OBSERVED DATA

24.0 months

GOVERNING BATCH

B2

MODEL SELECTED

Common slope, batch-specific intercepts

POOLABILITY ALPHA

0.25

CONFIDENCE BOUND

Lower one-sided 95% confidence bound on the mean response

EXTRAPOLATION

No. Supported period is within observed data.

Dataset

Attribute

assay

Unit

not recorded

Condition

25C/60RH

Direction

decreasing

Number of batches

Not recorded in metadata

Time points (months)

Not recorded in metadata

Row count used in fit

42

Required-column validation

All required columns present (assumed)

Model

Model kind

Common slope, batch-specific intercepts **common_slope_batch_intercepts**

Model formula

value ~ time + batch (common slope, batch-specific intercepts)

Estimated parameters

b0_B1

99.8623

b0_B2

98.9486

b0_B3

100.802

b1

-0.488847

Residual df

38

Residual standard error (s)

0.331206

Batches in fit

B1, B2, B3

Poolability

Decision

Partial pooling (common slope, batch-specific intercepts) **partial**

Alpha (significance level)

0.25 (ICH Q1E reference alpha = 0.25)

Guidance profile

Q1A_R2+Q1E

Step 1 p-value for slopes

$0.9056 \geq \alpha$, proceed to intercept test

Step 2 p-value for intercepts

$1.658e-16 < \alpha$, intercepts differ, common slope

Poolability notes

- step1 slopes: F-test on time:C(batch), $p=0.9056$ ($\alpha=0.25$)
- step2 intercepts: F-test on C(batch), $p=1.658e-16$ ($\alpha=0.25$)

Shelf Life Estimate

Supported shelf life

17 months

Statistical crossing (months)

18.0

Observed long-term data (months)

24.0

Confidence bound used

Lower one-sided 95% confidence bound on the mean response
(level = 0.95)

lower_one_sided_95_mean

Crossing status

Crossed specification within the evaluated horizon

crossed

Governing batch

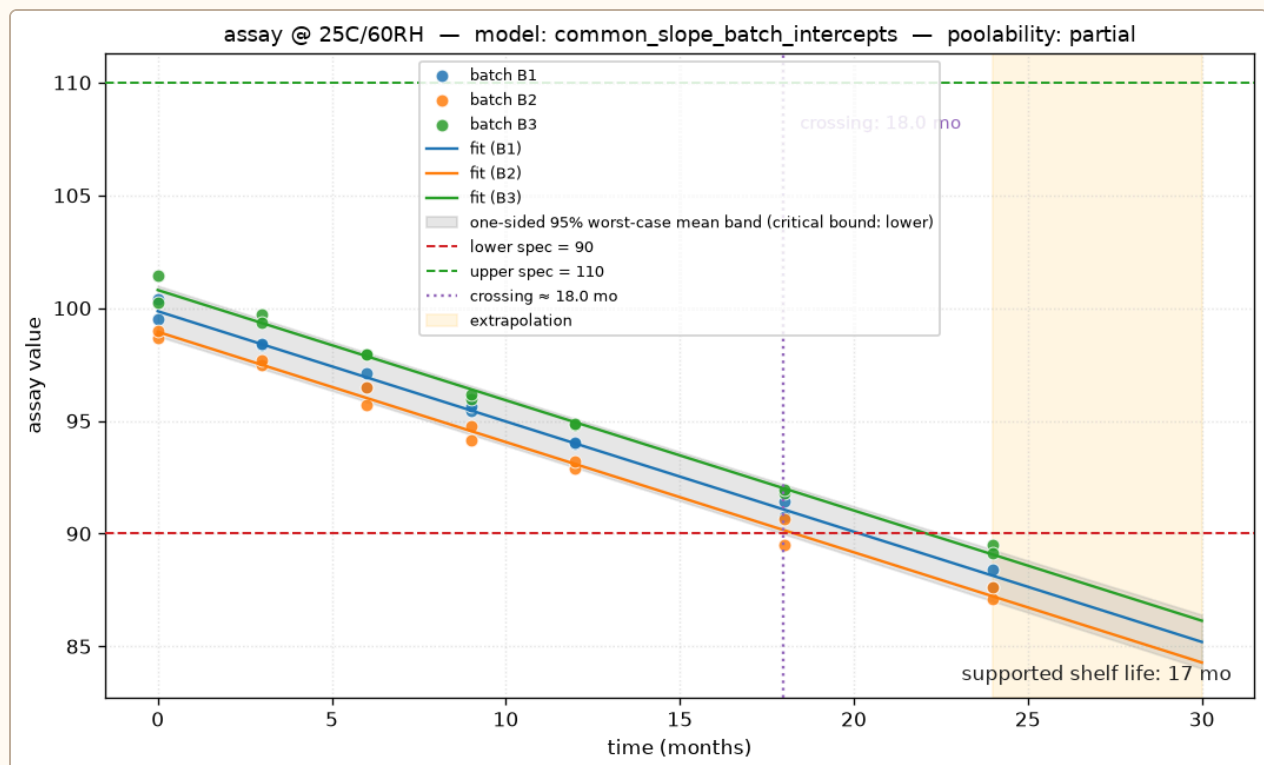
B2

Extrapolation

No. Supported period is within observed data.

The supported shelf life is rounded *down* to the nearest whole month, per ICH Q1E.

Confidence Bound Plot



Observed data by batch, fitted regression, the lower one-sided 95% confidence bound on the mean response, and the relevant specification limit. Crossing point and extrapolation region are marked. Plot file: confidence_plot.png.

Diagnostics and Assumptions

Linearity: ok

Homoscedasticity: ok

Normality of residuals: ok

Influential points (Cook's d): 4

Diagnostic notes

- Influence check: 4 observation(s) have Cook's distance $> 4.0/n = 0.0952$ (row indices [13, 24, 28, 29]). A single observation controls the fit. Consider a sensitivity analysis with and without these points; do not auto-exclude.

Warnings and Limitations

data_quality[info] duplicate_batch_time_no_replicate: 42 rows share (batch, time, attribute) ; no 'replicate' column

data_quality summary: 0 error(s), 0 warning(s), 1 info; can_analyze=True

step1 slopes: F-test on time:C(batch), $p=0.9056$ ($\alpha=0.25$)

step2 intercepts: F-test on C(batch), $p=1.658e-16$ ($\alpha=0.25$)

Influence check: 4 observation(s) have Cook's distance $> 4.0/n = 0.0952$ (row indices [13, 24, 28, 29]). A single observation controls the fit. Consider a sensitivity analysis

with and without these points; do not auto-exclude.

Significant-change assessment (ICH Q1A)

Accelerated condition

not recorded

Intermediate condition

not recorded

Extrapolation allowed

yes

Rationale

no accelerated data

Criteria fired

Criterion

First change (months)

Evidence

accelerated

not recorded

intermediate

not recorded

Model effects

Model

fixed (Q1E default fixed effect batch)

Reproducibility

Tool version

1.0.4

Analysis timestamp (ISO-8601)

2024-06-01T00:00:00Z

Input file SHA-256

65138b8343548cfcc775d822e124c556ed47e62fdec455bc4ccd7d946f8a497e

Random seed (if any)

none / deterministic core

Library versions

<code>python</code>
<code>3.12.10</code>
<code>pandas</code>
<code>3.0.3</code>
<code>numpy</code>
<code>2.4.6</code>
<code>scipy</code>
<code>1.17.1</code>
<code>statsmodels</code>
<code>0.14.6</code>
<code>matplotlib</code>
<code>3.11.0</code>
<code>jinja2</code>
<code>3.1.6</code>

Disclaimer. This report is ICH Q1E-inspired and intended for educational, exploratory, and reproducible decision-support use. It is not a substitute for qualified regulatory, statistical, or quality review. The toolkit does not provide 21 CFR Part 11 audit trails, electronic signatures, or data integrity controls, and is not a validated GxP system.

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